

James Bruning

Chicago, Illinois • 630-947-6916 • jimtbruning@gmail.com • github.com/jabrunin001 • linkedin.com/in/jtb96

SUMMARY

Senior Data Engineer with 6+ years building data platforms in higher education. I design batch and streaming pipelines in Python, PySpark, Airflow, and Azure, replace legacy vendor systems, and stand up the governance and automation that keep them running. Comfortable owning a problem from the Spark migration down to the stakeholders who depend on the numbers.

EXPERIENCE

Senior Data Engineer

Northwestern University, Chicago

2021 – Present

- Replaced a legacy vendor system with a containerized Airflow platform on Docker, cutting annual costs **5%** and giving the team something it could actually maintain.
- Built a university-wide data mart in Azure SQL Server, integrating **120+** sources through Azure Data Factory so three business units could run their own analytics instead of filing requests.
- Built a Python and ML forecasting model for **\$2M+** in staff-salary budgeting, raising forecast accuracy from **73% to 97%**.
- Migrated **20+** ETL pipelines from SQL to PySpark and Spark SQL, dropping large-batch runtimes from **35 minutes to under 7**.
- Set up the team's dbt and Snowflake analytics-engineering practice: reusable models and CI/CD that made new pipelines faster to ship and easier to onboard onto.
- Replaced a 31-step monthly reporting process (Cognos, Excel, Smartsheet, and 80+ Tableau Prep flows) with one configurable Python script, cutting it from **3 hours to 15 minutes**.
- Modeled a cross-office data layer linking IRB, IACUC, IBC, and Sponsored Research on a shared project ID, which gave the office its first integrated view of research activity.
- Automated recurring Cognos report generation (GA101/GA201) with Playwright, writing output straight to SharePoint and removing hours of manual work each cycle.
- Co-authored the Tableau Server governance SOP covering site structure, naming, access control, and publishing standards, now used across the Office for Research.
- Led the organization's first formal data-governance program on a 90-day mandate from the Provost: a working data dictionary, named stewards per office, and a two-tier committee structure.
- Built a request-analysis tool in React and the Anthropic API that turns free-form stakeholder emails into structured action items, risks, and open questions.

Data Analyst

Northwestern University, School of Communication

2019 – 2021

- Built an automated validation pipeline to audit and reconcile **\$20M** in grant spending, cutting processing time **65%** and raising accuracy to **99%**. It became the basis for the office's compliance reporting.
- Built reusable data models from **20+** sources behind the BI dashboards used for program evaluation and budgeting; graduate-student satisfaction rose from **7.3 to 8.8** over that period.
- Migrated the grants database and built a self-service query platform that cut ad-hoc reporting requests by **300%**.

EDUCATION

M.S. Data Science, Northwestern University (2022–2024)

B.A. Physics, Indiana University (2014–2018)

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Spark SQL

Platforms & warehouses: Snowflake, Azure (Data Factory, SQL Server), Databricks

Engineering: Airflow, dbt, Docker, Apache Iceberg, Trino, Playwright

Visualization: Tableau (Desktop, Server, Prep), Power BI

AI: Anthropic API (Claude), React